

Microchips could be argued as the most sophisticated thing that humanity has ever created. They embody a culmination of knowledge from fields such as chemistry, quantum physics, mechanics and even optics. They power CPUs, rockets, fridges and pretty much anything else you could imagine. The semiconductor industry is worth more than a trillion dollars yet microchip manufacturers have little to no competition. This is due to the high complexity of the manufacturing process.

Firstly, quartz sand is heated, distilled multiple times and then crystallised. This yields a 99.99% silicon 'ingot' which is then cut and polished into silicon wafers with a diameter of 30cm. These wafers undergo various chemical processes. To begin with, the wafer is coated in a layer of a solution called photoresist. Areas of the photoresist exposed to ultraviolet light lose integrity and become more soluble which allows companies to print designs using ultraviolet light.

The ultraviolet light comes from an EUV lithography machine. These machines are not particularly cheap. They cost \$300 Million to make and only one company manufactures them, ASML. Underneath the machine, a high power laser fires two beams of light that are amplified and then reflected off of mirrors into a source vessel. In the source vessel, the first beam of light collides with a tin droplet and flattens it out. The second - higher powered - beam vaporises the tin, turning it into glowing plasma. Electrons in the tin atoms are excited from the 4D to the 4F orbital. Once they are de-excited, they release 13nm EUV light. This wavelength does not occur naturally on earth, and the selection of tin is due to its emission spectrum having a clearly distinguished peak at 13nm.

The tin is stored as a liquid in a storage tank containing high pressure nitrogen which allows it to be pushed out. It is fed through a microscopic nozzle which is squeezed by a Piezo Electric Transducer. The tin droplet is shot out at 100m/s. High speed cameras allow for the calculation of the trajectory of the droplets. This data is received by the actuated mirrors which align the pulses of light to hit the droplets. The process of shooting these tin droplets occurs 50,000 times per second. All the EUV light is collected by a spherical mirror called a collector and focused through a set of mirrors made of alternating layers of silicon and molybdenum. These mirrors need to be manufactured with very tight tolerances, with deviations of the surface being less than an atom thick. When the uniform, circular beam of EUV light is sent through these mirrors, the light can be transformed to various shapes depending on the stencil being used. This can be done as some of the mirrors are composed of hundreds of independently controlled smaller mirrors.

The light reflects off the stencil and onto the silicon wafer. This stencil is a premade microchip design that the company prints for an integrated circuit. The resolution of what can be projected from a stencil and onto a wafer is as low as 10nm. For reference, a carbon atom is around 0.15nm large. Hence, billions of transistors can fit into a CPU less than the size of your hand. An entire silicon wafer can be covered in 1000s of identical stencil patterns in less than half a minute and after it is covered, the wafer undergoes etching.

During the process of etching, the wafer is cooked, developed and then the soluble resist layer is washed away to reveal a 3D pattern of the stencil design. The semiconductive property of silicon means it is ideal for ion doping. The wafer is doped with gallium or arsenic depending on whether the intended doping is p-type or n-type. This doping can then form transistors allowing for the creation of integrated circuits.

The process of lithography, etching and ion implantation is repeated dozens of times on one wafer. From start to finish, one silicon wafer can be on the production line for 3 months, with 100s of processes being performed to manufacture it. This is due to the large amount of microscopic layers that the microchips need for wiring of transistors.

Due to impurities, some cores on a CPU may not work. This could be due to dust or other faults. Intel uses a classification system which allows customers to know how many working cores the CPU has. To minimise impurities, the microchip manufacturing process is done in a cleanroom, with little human interaction occurring. The wafers are carried in airtight boxes attached to rails on the roof of the manufacturing floor.